Saumyen Kundu Ph.D.

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Date of Birth: 13th July, 1991

Research Interest

My research focuses on the collider phenomenology of beyond the Standard Model (BSM) physics and its connection to cosmology, particularly in the context of dark matter. I am interested in developing and analyzing models with rich phenomenology across colliders and cosmological probes, using tools such as jet substructure techniques and machine learning methods (e.g., BDTs, MLPs). My broader interests include:

- Beyond Standard Model (BSM) Physics
- Dark Matter Search
- Collider Phenomenology
- Applications of Machine Learning
- Cosmology and Astroparticle Physics

Work Experience

Apr, 2024– *present*	*Post-Doctoral Fellow* Harish-Chandra Research Institute Prayagraj, Uttar Pradesh, India
Sep, 2015– *May*, 2017	*Assistant Professor*, Department of Physics Seacom Skills University, Sriniketan, West Bengal

Teaching Experience

Jan, 2025 – *May*, 2025	*Teaching Assistant*, Harish-Chandra Research Institute, Prayagraj, Uttar Pradesh • *Particle Physics* course for M.Sc. students.
Aug 2017 – *Dec* 2022	*Teaching Assistant*, BITS Pilani, K.K. Birla Goa Campus, Zuarinagar, Goa • *Particle Physics* course for M.Sc. students. • *Laboratory Physics* course for B.E. students.
Sep 2015 – *May* 2017	*Assistant Professor*, Seacom Skills University, Sriniketan, West Bengal • *Physics (Honours)* for undergraduate students. • *Engineering Physics* and *Electromagnetic Field Theory* for B.Tech. students.

Education

2017 – 2023	**Ph.D.** in High Energy Physics **Advisor:** Prof. Prasanta Kumar Das **BITS Pilani, K.K. Birla Goa Campus**, Goa
2013 – 2015	**M.Sc.** in Physics (specialization in *Particle Physics*) **Visva-Bharati**, Santiniketan, West Bengal
2010 – 2013	**B.Sc.** in Physics (Honours) **Visva-Bharati**, Santiniketan, West Bengal

Technical Skills

- **Software packages:** CERN-ROOT, TMVA, Mathematica, FeynRules, LanHEP, MadGraph5, Whizard, CalcHEP, Pythia8, Delphes3, FastJet3, MadDM, MicrOMEGAs, MadAnalysis5
- **Languges:** C++, Python, bash-script.

Publications

Articles

- *Tracing Neutrino Non-Standard Interactions through Charged Lepton Collisions*
  Sudip Jana, **Saumyen Kundu**, Santosh Kumar Rai
  under review in Phys. Rev. [arXiv:2507.10703].
- *Revisiting the inert scalar dark matter with vector-like quarks*
  Prasanta Kumar Das, Shyamashish Dey, **Saumyen Kundu**, Santosh Kumar Rai
  JHEP **05** (2025) 027 [arXiv:2412.17719].
- *Exploring the electromagnetically interacting dark matter at the International Linear Collider*
  Manish Kumar Sharma, **Saumyen Kundu**, Prasanta Kumar Das
  under review in Phys. Rev. D [arXiv:2301.02514].
- *Jet substructure probe to unfold singlet-doublet dark matter in the presence of non-standard cosmology*
  Prasanta Kumar Das, Partha Konar, **Saumyen Kundu**, Sudipta Show
  JHEP **06** (2023) 198 [arXiv:2301.02514].
- *SN1987A cooling due to Plasmon-Plasmon scattering in the Randall-Sundrum Model*
  Manish Sharma, **Saumyen Kundu**, Prasanta Kumar Das
  Phys. Dark Univ. **40** (2023) 101218 [arXiv:2304.00380].
- *EFT analysis of leptophilic dark matter at future electron-positron colliders in the mono-photon and mono-Z channels*
  Saumyen Kundu, Atanu Guha, Prasanta Kumar Das, P.S. Bhupal Dev
  Phys. Rev. D **107** (2023) 015003 [arXiv:2110.06903].

Conference Proceedings

- *Jet substructure probe to freeze-in dark matter in alternative cosmological background*
  Saumyen Kundu, Sudipta Show, Partha Konar, Prasanta Kumar Das
  Eur. Phys. J. Spec. Top. **233**, 2129–2134 (2024).
- *Freeze-In Dark Matter Search at Colliders with Jet Substructure Analysis*
  Saumyen Kundu, Sudipta Show, Partha Konar, Prasanta Kumar Das
  Springer Proc. Phys. **304** (2024) 393-396.
  Contribution to: 25th DAE-BRNS High Energy Physics Symposium.

Reports

- *The International Linear Collider: Report to Snowmass 2021*
  ILC International Development Team
  Contribution to: Snowmass 2021 e-Print: 2203.07622 [physics.acc-ph].

Recent Talks and Presentations

19 – 23 *December*, 2024
Enhancing Dark Matter Phenomenology with Vector-Like Quarks in the IDM
XXVI DAE-BRNS HEP Symposium, Banaras Hindu University, India
Poster Presentation

10 – 12 *March*, 2023
Jet substructure probe to freeze-in dark matter in alternative cosmological background
Frontiers in Particle Physics, 2023, CHEP, IISc, Bangalore, India
Poster Presentation

12 *December*, 2022	*Freeze-in Dark Matter Search at Colliders with Jet Substructure Analysis* **XXV DAE-BRNS HEP Symposium**, IISER Mohali, India **Oral Presentation**
07 *December*, 2021	*Leptophilic Dark Matter at International Linear Collider* **AstroDark-2021**, Kavli IMPU, Japan **Poster Presentation**
11 *November*, 2021	*Searching for Dark Matter signatures at the ILC with Leptophilic Effective Models* **Anomalies 2021**, IIT Hyderabad, India **Oral Talk**
04 *November*, 2021	*Searching for Leptophillic Dark Matter in Linear Colliders* **Brookhaven Forum 2021**, Brookhaven National Laboratory, USA **Oral Talk**
27 *October*, 2021	*Quest for Dark Matter at International Linear Collider* **ILC Workshop on Potential Experiments (ILCX2021)** **Oral Talk**
25 *January*, 2020	*Searching for Leptophillic Dark Matter in Linear Colliders* **Goa Physics Meet 2020**, IIT Goa, Ponda, Goa **Poster Presentation**

Schools/Workshops/Conferences Attended

14 *November*– 18 *November*, 2021	• *17th International Conference on Interconnections between Particle Physics and Cosmology (PPC 2025)* IIT Hyderabad, India
13 *March*– 24 *March*, 2021	• *Less Travelled Path to the Dark Universe, 2023* TIFR-ICTS, Bengaluru, India
29 *November*– 2 *December*, 2021	• *Kashiwa Dark Matter Symposium, 2021* ICRR, University of Tokyo, Japan Online
22 *November*– 26 *November*, 2021	• *Computational Tools for High Energy Physics and Cosmology* Institut de Physique des 2 Infinis de Lyon (IP2I), Lyon, France Online
22 *November*– 26 *November*, 2021	• *Hunting SUSY @ HL-LHC, 2021* ICTS-TIFR, Bengaluru, India Online
18 *October*– 22 *October*, 2021	• *Higgs 2021* Stony Brook University, Simons Center for Geometry and Physics Online
23 *August*– 4 *September*, 2021	• *CERN-Fermilab Hadron Collider Physics Summer School, 2021* CERN, Switzerland Online
28 *July*– 6 *August*, 2020	• *International Conference on High Energy Physics, 2020* Prague, Virtual Conference
18 – 22 *November*, 2019	• *MadGraph School 2019* Institute of Mathematical Science, Chennai, Tamil Nadu
14 *October*–, 9 *November*, 2019	• *SERB Preparatory School in Theoretical High Energy Physics* Tezpur University, Tezpur, Assam
3 – 5 *February*, 2019	• International Conference on "*Recent Issues in Nuclear and Particle Physics*" Visva-Bharati, Santiniketan, West Bengal

References

- *Prof. Prasanta Kumar Das*
  Professor
  Department of Physics
  BITS Pilani, K.K. Birla Goa Campus, Goa, India
  pdas@goa.bits-pilani.ac.in

- *Prof. Santosh Kumar Rai*
  Professor
  Harish-Chandra Research Institute,
  Prayagraj, Uttar Pradesh, India
  skrai@hri.res.in

- *Prof. Partha Konar*
  Professor
  Theoretical Physics Division,
  Physical Research Laboratory, Ahmedabad, India
  konar@prl.res.in

- *Prof. P.S. Bhupal Dev*
  Associate Professor
  Department of Physics and McDonnell Center for the Space Sciences,
  Washington University, St. Louis, Missouri, USA
  bdev@wustl.edu

- *Dr. Sudip Jana*
  Assistant Professor
  Harish-Chandra Research Institute,
  Prayagraj, Uttar Pradesh, India
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